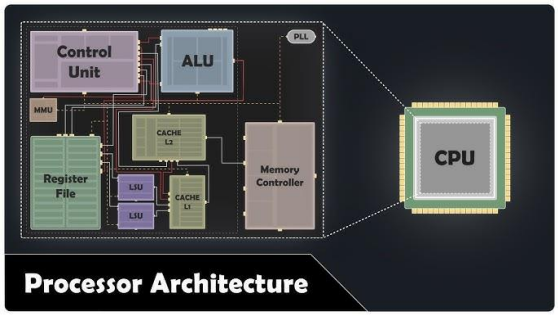
Cycle-1

Experiment-1

Q: Familiarization to computer hardware such as CPU, Memory, Motherboard etc.

Aim: To study and understand the basic concepts of computer hardware parts.

1. **CPU (Central Processing Unit)**



The CPU is the brain of the computer. It performs all processing, calculations, and controls all the operation of the system.

It consists of:

- ALU (Arithmetic Logical Unit): Performs arithmetic and logical operations.
- CU (Control Unit): Controls the execution of the instructions.
- Registers: Small memory units for temporary data storage.

2. **Memory (RAM-Random access memory)**

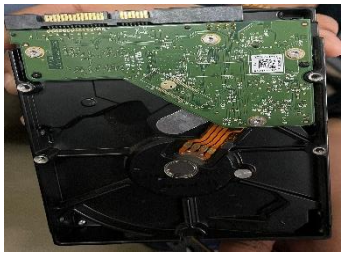
Ram is the temporary memory used to store data while programs are running. It increases system speed and performance. Data lost when power is turned off.

3. Motherboard

The motherboard is the main circuit bord. All the components like CPU, RAM, Storage and peripherals are connected to it.

4. Storage Devices

i. HDD (Hard Disk Drive)



Stored data permanently, slower speed**.**

ii. SSD (Solid State Drive)

Faster storage device used in modern systems.

5. **Power Supply Unit (PSU)**

Supplies electrical power to all the internal components.

6. **GPU (Graphics Processing unit)**

Used for graphics and display processing. Helps in gaming, video editing and graphics-heavy applications